

MW 1

$$1 \text{ sec} = 1,000,000 \mu\text{s} \quad 60 \text{ sec} = 1 \text{ min} \quad 60 \text{ mins} = 1 \text{ hour} \quad 24 \text{ hours} = 1 \text{ day}$$

Problem 1)

$$n = 1000 \quad n = 10,000 \quad n = 100,000 \quad n = 1,000,000$$

$$B \ n^2 \quad 1000^2 \mu\text{s} \quad 10,000^2 \mu\text{s} \quad 100,000^2 \mu\text{s} \quad 1,000,000^2 \mu\text{s}$$

$$= 1 \text{ sec.} \quad = 1.67 \text{ mins.} \quad = 2.28 \text{ hrs.} \quad = 11.57 \text{ days}$$

$$C \ n^3 \quad 1000^3 \mu\text{s} \quad 10,000^3 \mu\text{s} \quad 100,000^3 \mu\text{s} \quad 1,000,000^3 \mu\text{s}$$

$$= 16.67 \text{ mins} \quad = 11.57 \text{ day} \quad = 31.71 \text{ years} \quad = 31,709.79 \text{ years}$$

$$\text{One-sec} = 1,000,000 \mu\text{s}$$

$$\text{One-min} = 60,000,000 \mu\text{s}$$

$$\text{One-hour} = 360,000,000 \mu\text{s}$$

$$\text{One-day} = 8,640,000,000 \mu\text{s}$$

$$\text{One-year} = 31,536,000,000 \mu\text{s}$$

divide the number
by the largest number
that it is greater than
one that will be the
unit

Note accounting for
leap years!!!

$$n = 1000$$

$$n = 10,000 \quad 100,000$$

$$n = 0.000,0001$$

$$0.001$$

6

$$0.01$$

$$0.1$$

$$1,000,000 / \text{sec}$$

$$100,000,000 / \text{min}$$

$$10,000,000,000 / \text{hour}$$

$$1,000,000,000,000 / \text{day}$$

$$1,000,000,000 / \text{mins}$$

$$1,000,000,000,000 / \text{day}$$

$$1,000,000,000,000,000 / \text{years}$$

$$1,000,000,000,000,000,000 / \text{years}$$

Problem 2)

a) $\Theta(n^2)$

line 2: $n - 2025 = n$

line 3: $\frac{n(n-1)}{2} = n^2$

b) $\Theta(n^{14})$

line 2: n^{14}

c) $\Theta(n^2)$

first part: n^2

second part: n

$n^2 + n = n^2$

d) $\Theta(n^2)$

first part: n^2

second part: n

$n^2 + n = n^2$

e) $\Theta(n^2)$

first item in list: $\Theta(1)$

second item in list: $\Theta(n)$

third item in list: $\Theta(n)$

fourth item in list: $\Theta(n^2)$

f) $\Theta(n^2)$

first loop: $\Theta(1)$

second loop: $\Theta(1)$

third loop: $\Theta(n^2) + \Theta(n) = \Theta(n^2)$

Problem 3 outp $\log_3 n + 1$ bc lines 1-3

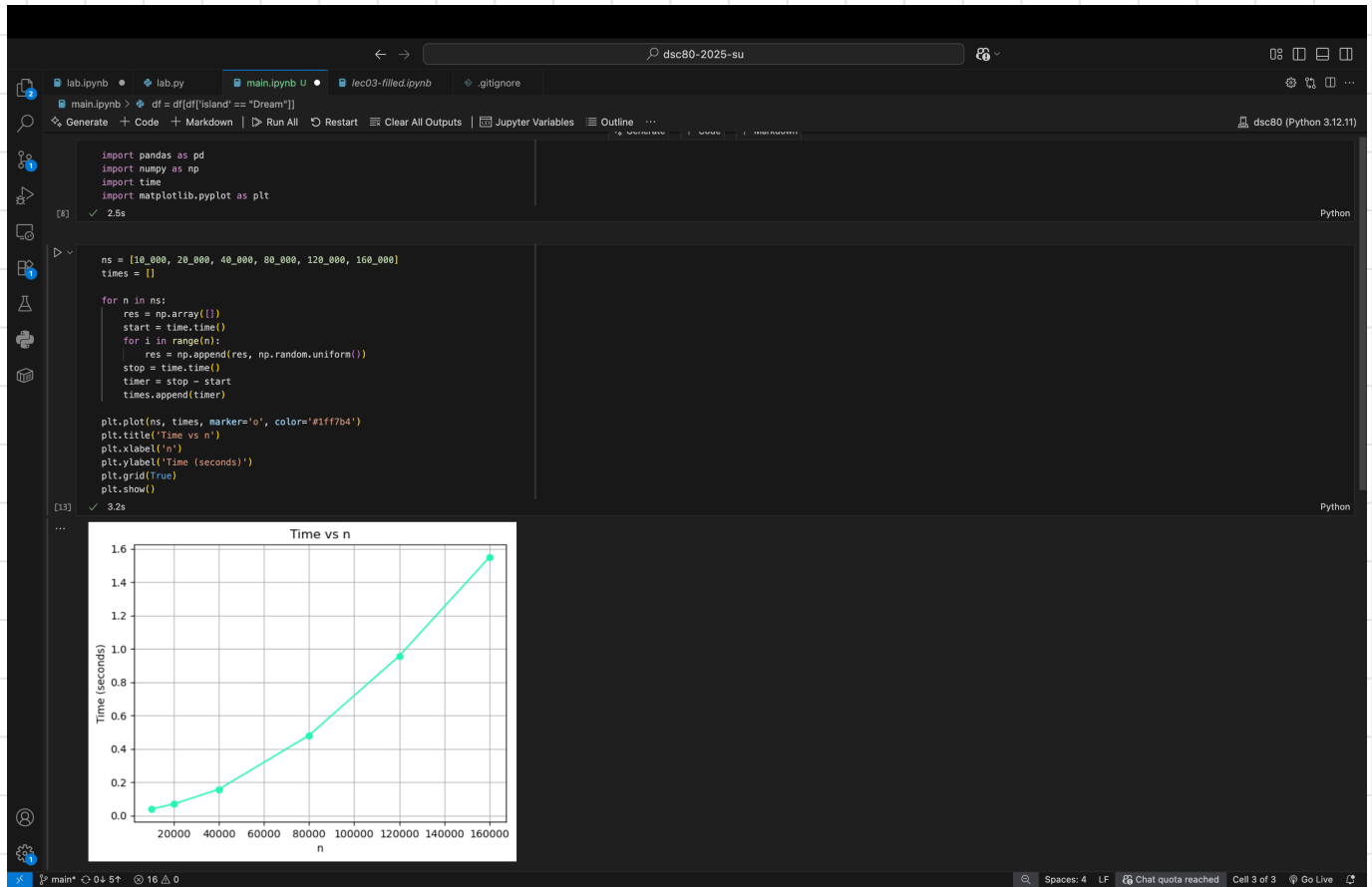
$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\text{inner} = \sum_{k=0}^n r^k = \frac{a(r^n - 1)}{r - 1} = \frac{3^{\log_3 n + 1} - 1}{3 - 1} = \frac{3 \cdot 3^{\log_3 n} - 1}{2}$$

$$= \frac{3^{n+1} - 1}{2} = \Theta(n)$$

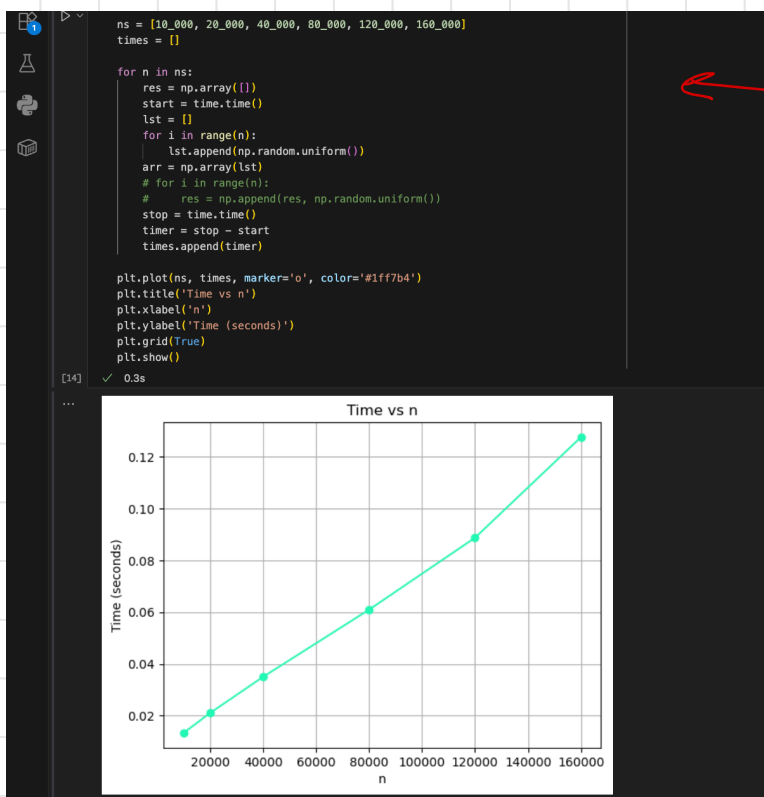
Problem 4 a) n^2

b)



c) $\Theta(n^2)$, this is $\Theta(n^2)$ because the For loop is $\Theta(n)$ and the `np.append` must be $\Theta(n)$ also which is repeated n times so the total complexity is $\Theta(n^2)$

d)



← all imports are above

Problem 5) a) This computes the largest i which squared is less than n

b) $\Theta(\sqrt{n})$ bc the loop will run $\frac{1}{2}n$ $i^2 \geq n$

$$\sqrt{i^2} \geq \sqrt{n} \quad \therefore i \geq \sqrt{n}$$